SRM TRP ENGINEERING COLLEGE
TIRUCHIRAPPALLI

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

ASSIGNMENT–2: MINOR PROJECT

Machine Learning Model Development and Performance Analysis

Course: AD25C01 – Exploratory Data Analysis

Project Topic: Online Shopping Purchase Prediction

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Team: Team 6

Academic Year: 2026–2027

Prepared using the cleaned dataset supplied for Assignment–1.

1. Abstract

This project develops and compares three supervised machine-learning classifiers to predict whether an online browsing session generates revenue. The supplied cleaned dataset contains 9,864 records and 27 columns, including the target variable Revenue. Logistic Regression, Decision Tree, and Random Forest are trained using a stratified 80:20 train-test split. Performance is measured with accuracy, precision, recall, F1-score, and ROC-AUC. Since purchase sessions are the minority class, the analysis emphasizes F1-score and ROC-AUC rather than relying on accuracy alone.

2. Problem Definition

Problem: Predict a binary outcome for a browsing session: Revenue = 1 (purchase/revenue generated) or Revenue = 0 (no purchase/revenue generated). **Target variable:** Revenue. **Feature variables:** all other 26 columns, including page counts and durations, bounce/exit rates, page values, special-day indicator, operating system, browser, region, traffic type, weekend, month indicators, and visitor-type indicators.

3. Dataset and Preparation

The Excel workbook includes Cleaned Data, Original Data, and Data Quality Report sheets. The cleaned sheet has 9,864 rows and 27 columns. The quality report records zero missing cells and zero duplicate rows after cleaning. Existing numeric scaling and encoded values were preserved; many fields contain standardized negative values and should not be interpreted as raw counts or durations. The target distribution is 8,288 sessions with Revenue = 0 and 1,576 sessions with Revenue = 1.

The data was divided into 80% training and 20% testing data using stratification and random_state=42. This produced 7,891 training rows and 1,973 testing rows. Class weighting was applied in the classifiers to account for the minority purchase class.

4. Models Developed

Logistic Regression: linear baseline classifier with standardization and balanced class weights. **Decision Tree:** tree-based classifier with depth and leaf-size constraints to reduce overfitting. **Random Forest:** ensemble of 250 trees, using balanced subsampling of classes.

5. Evaluation Metrics

Accuracy: proportion of all predictions that are correct. **Precision:** proportion of predicted purchase sessions that are actual purchase sessions. **Recall:** proportion of actual purchase sessions correctly identified. **F1-score:** harmonic mean of precision and recall. **ROC-AUC:** threshold-independent measure of ranking/discrimination ability. A confusion matrix summarizes true and false predictions.

6. Model Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.7182	0.2995	0.5714	0.3930	0.7223
Decision Tree	0.6107	0.2382	0.6540	0.3492	0.6634
Random Forest	0.8363	0.3889	0.0444	0.0798	0.7310

The best model by F1-score, with ROC-AUC used as a tie-breaker, is **Logistic Regression**. It achieves F1-score 0.3930 and ROC-AUC 0.7223. Model choice should be considered in the context of the application: high recall may be important when missing likely purchases is costly, while high precision may be preferred when outreach costs are high.

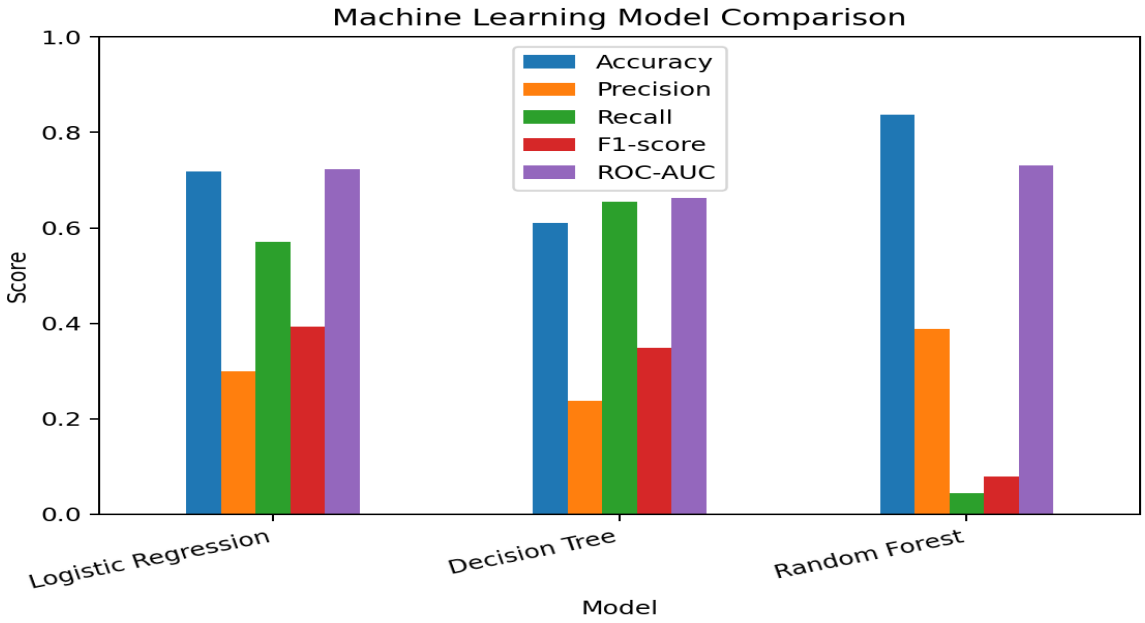


Figure 1. Test-set performance comparison across the three classifiers.

7. Confusion Matrix and ROC Curve

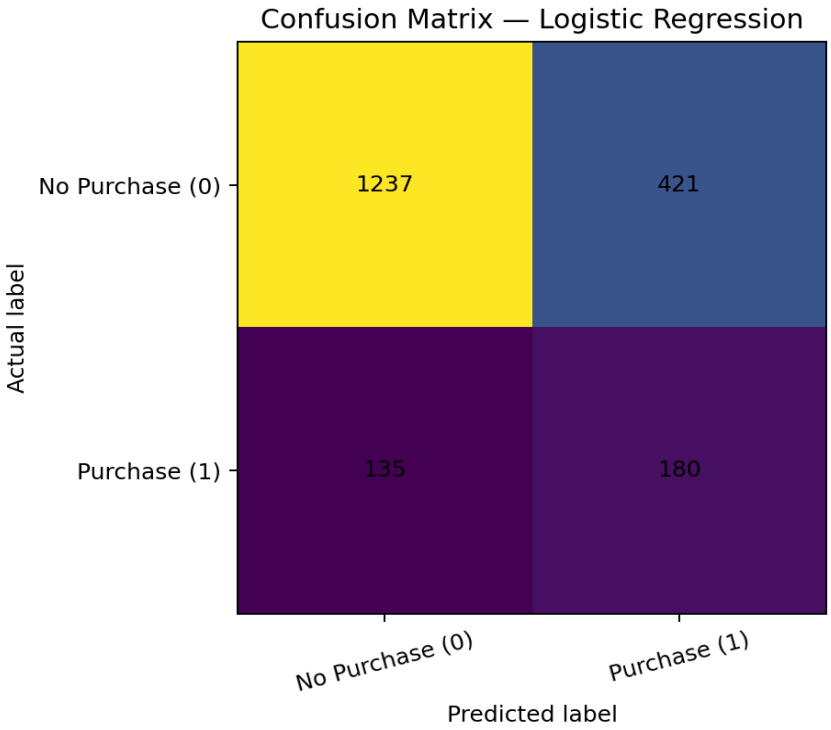


Figure 2. Confusion matrix for Logistic Regression on the held-out test set.

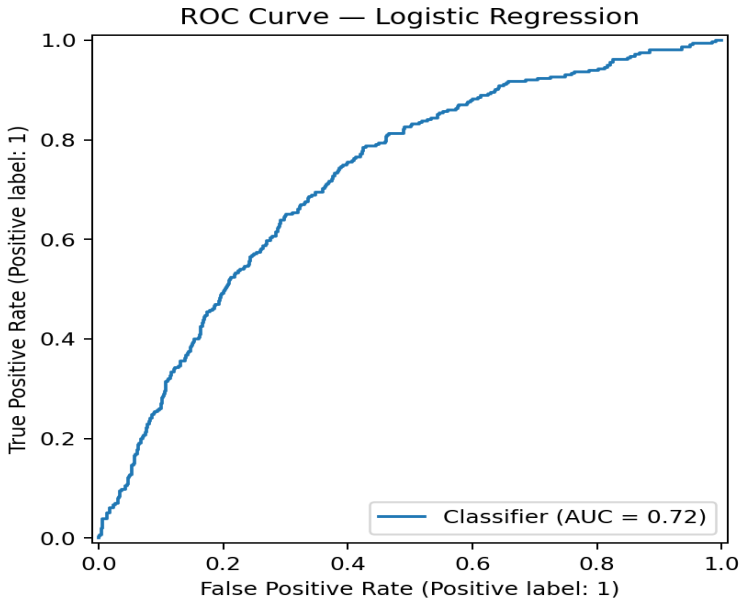


Figure 3. ROC curve for Logistic Regression; test ROC-AUC = 0.7223.

8. Feature Importance

Permutation importance was calculated on the test set using F1-score. It measures the average change in F1-score when each feature is shuffled. The strongest feature in this analysis is **ProductRelated_Duration**, with mean F1 decrease 0.1766. This suggests the fitted model relies on this feature for its test-set predictions; it does not establish that the feature causes a purchase.

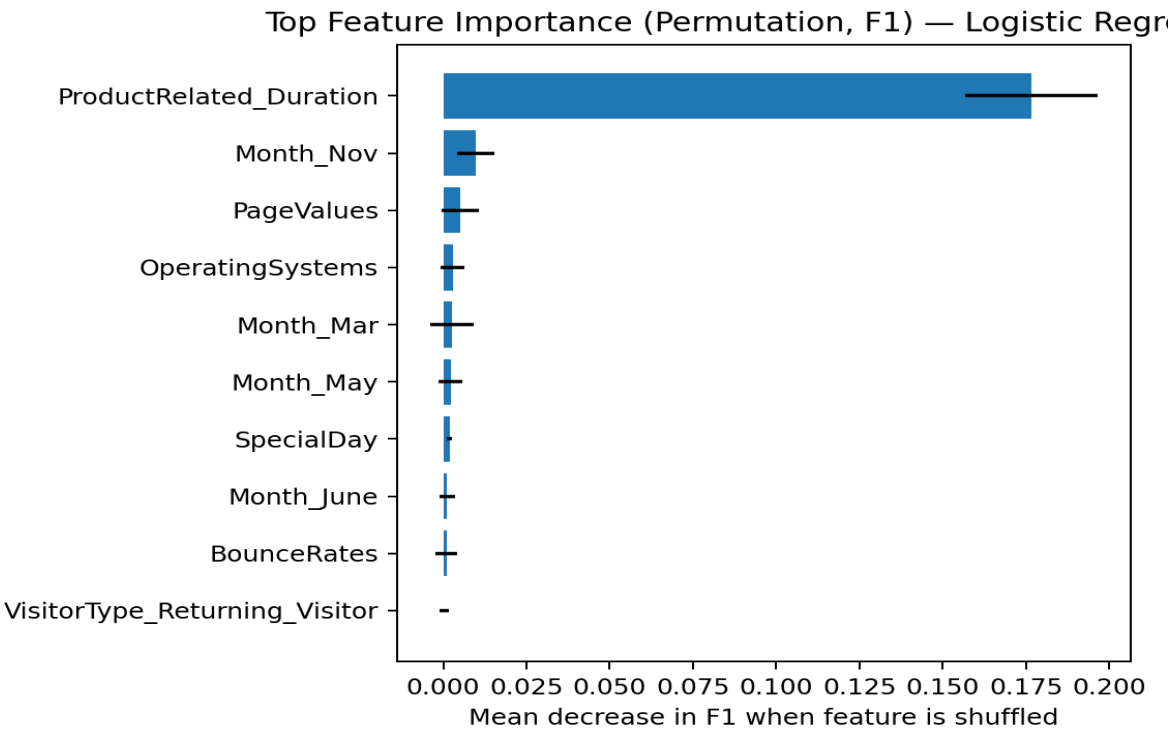


Figure 4. Top ten features ranked by permutation importance.

9. Limitations

1. The target is imbalanced: about 15.97% of records have Revenue = 1. 2. The supplied cleaned file contains already scaled and encoded values, so the original units and categorical labels may not be recoverable from this sheet alone. 3. A single train-test split gives one estimate and can vary with the split. 4. No external or future-period dataset was supplied, so generalization to other websites or time periods has not been established. 5. Model results depend on the selected hyperparameters and classification threshold.

10. Future Improvements

Use stratified cross-validation and hyperparameter search; tune the decision threshold according to business costs; compare class-weighting with resampling performed only on training folds; calibrate predicted probabilities; check drift across time periods; and validate on a newer independent dataset. If raw data is available, document the meanings and units of all features and reproduce the cleaning and encoding steps in a transparent pipeline.

11. Real-World Applications

A validated model of this type could support e-commerce analytics by identifying sessions with higher purchase likelihood, guiding website usability investigations, informing opt-in marketing prioritization, and helping teams understand which browsing behaviours are associated with revenue. Predictions should support, not replace, human judgement and should be monitored for bias and performance drift.

12. Conclusion

Three classification models were developed and evaluated using the supplied cleaned dataset. By the selected criterion (F1-score, followed by ROC-AUC), Logistic Regression is the recommended model among the tested candidates. The final selection should be revisited if the practical goal changes—for example, if maximizing recall is more important than balancing precision and recall. The notebook and saved model files allow the analysis to be reproduced.

References

1. Scikit-learn documentation: model evaluation metrics, LogisticRegression, DecisionTreeClassifier, RandomForestClassifier, and permutation importance. https://scikit-learn.org/stable/

2. Course Assignment Instructions with Rubrics, AD25C01 – Exploratory Data Analysis, Assignment–2, supplied by the student.